A Hyperspectral Anomaly Detection Method Based on Low-Rank and Sparse Decomposition With Density Peak Guided Collaborative Representation

Shou Feng, *Member, IEEE*, Shulu Tang, Chunhui Zhao, and Ying Cui

***Abstract*— The low-rank and sparse decomposition model (LSDM) has been widely studied by researchers and has successfully solved the problem of hyperspectral image (HSI) anomaly detection (AD). The traditional LSDM usually ignores the information of the low-rank matrix, which only detects the anomalous targets by using the sparse component. To utilize both the sparse component and the low-rank component comprehensively, an anomaly detector for HSIs based on LSDM with density peak guided collaborative representation (LSDDPCRD) is proposed in this article. First, the LSDM technique with the mixture of Gaussian model is used to decompose the original HSI, which can also alleviate the background noise contamination problem. Then, the low-rank matrix is detected by the density peak guided collaborative representation detection algorithm, while the sparse matrix is calculated according to the Manhattan distance. In addition, an entropy-based adaptive fusing method is designed to combine the results obtained from the low-rank matrix and the sparse component. It could choose the fusing weights adaptively according to the characteristics of an HSI. The experimental results indicate that the LSDDPCRD performs better than eight classical and state-of-the-art AD algorithms (GRX, LRX, SRX-Segmented, CRD, RPCA-RX, LSMAD, LRASR, and LSDM-MoG) on four real HSIs.**

***Index Terms*— Collaborative representation, density peak, hyperspectral anomaly detection (AD), hyperspectral image (HSI), low-rank and sparse decomposition.**

I. Introduction

COMPARED with general optical images, hyperspectral images (HSIs) usually have many spectral bands with very rich spectral information, and they also contain spatial information of land-covers [1], [2]. In recent years, how to take advantage of the characteristics of HSIs to detect targets has been widely studied [3], [4]. Hyperspectral target detection technology includes particular target detection if prior information of targets is available and anomalous target detection [5], [6]. Because of the difficulty in obtaining prior information, the anomaly detection (AD) is very useful in many applications [7]–[9]. The so-called "anomaly" means that the spectral or spatial information of the anomalous target is very different when compared with the background [10], and the percentage of the anomalies in the HSI is very small [11].

The aim of hyperspectral AD is to split the anomalous targets apart from pixels belonging to the background [12]. The most classical algorithm for hyperspectral AD is the RX method, which is served as a benchmark [13]. Assuming that the background pixels are normally distributed, the procedure of the RX algorithm is to figure out the statistical distribution of the background by taking advantage of the whole image [14]. This algorithm is also called the global RX detector (GRXD). However, using the statistical information of the whole image is too simple to estimate the distribution of the background, which can reduce the detection accuracy [13]. Therefore, the local calculation is used instead of global calculation, and that is local RX (LRXD) [15]. Furthermore, several detectors have been proposed as improved versions of the RX, such as the kernel RX (KRX) algorithm [16], multiwindow RX algorithm [17], [18], and the SRX-Segmented algorithm [19].

However, the assumption of RX-based methods may be not very consistent with the real data distribution of HSIs. Therefore, other types of AD methods are designed. With the development of compression sensing, many scholars pay more attention to the representation theory [5]. For instance, the basic idea of the collaborative representation-based detector (CRD) introduced in [20] is that every background pixel can be rebuilt by several pixels in its surrounding field, but anomalous targets cannot [21]. The AD methods mentioned above only take the spectral information of an HSI into consideration, but ignore the spatial information of HSIs [22]. This phenomenon leads to the waste of hyperspectral data to a certain extent, and the detection efficiency of the detection method has not been fully developed. Through a large number of experiments, researchers have verified the effectiveness of

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The authors are with the College of Information and Communication Engineering, Harbin Engineering University, Harbin 150001, China, and also with the Key Laboratory of Advanced Marine Communication and Information Technology, Ministry of Industry and Information Technology, Harbin Engineering University, Harbin 150001, China (e-mail: fengshou@hrbeu.edu.cn; tangshulu@hrbeu.edu.cn; zhaochunhui@hrbeu.edu.cn; cuiying@hrbeu.edu.cn).

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using both spatial and spectral information to improve detection efficiency [23], [24]. To solve this problem, researchers have also put forward a variety of detection algorithms by utilizing the spatial and spectral joint information and characteristics, such as the local summation AD method (LSAD) [25] and the local sparsity divergence (LSD) [26].

Recently, the idea of low-rank and sparse decomposition is very popular, and this technique has been used to solve the task of hyperspectral AD successfully, which optimizes the detection performance to some extent [27]. The low-rank and sparse matrix decomposition model (LSDM) assumes that the HSI can be decomposed into two components, which are the low-rank component and sparse anomaly component [28]–[30]. Then, the Euclidean distance of pixels is calculated for similarity measurement on the sparse component. Normally, robust principal component analysis (RPCA) is taken to achieve LSDM [31]–[34], and then the spare component is detected by the RX algorithm. This detector is called RPCA-RX [28]. A low-rank and sparse representation-based detector (LRASR) was described in [21] to detect anomalies. The detector assumed that a dictionary which consists of pixels of all background classes can be used to reconstruct background pixels. For the LSMAD method proposed in [35], the Mahalanobis distance was employed to replace the Euclidean distance for similarity measurement, which further improves the detection accuracy. However, most of the traditional LSDM method and its improved versions mentioned above only deal with the sparse parts, but ignore the low-rank parts. In fact, the information of some anomalies and noise which have not been separated may still remain in the low-rank part. Therefore, it is necessary to further detect the low-rank part while making use of the sparse part. In this article, a detector based on LSDM with density peak guided collaborative representation (LSDDPCRD) is proposed for HSIs. The major contribution of this article includes the following.

First, the LSDDPCRD is based on the LSDM theory combined with the mixture of Gaussian (MoG) model. Because of the real environment and situation of hyperspectral imaging, different types of noise widely exist in an HSI. The mixture noise model can tackle with different types of noise, which is more in line with the actual complex data distribution of HSIs. By taking advantage of this, the proposed method can get more accurate results, as the noise problem has less influence on the detection results.

Second, different from the traditional LSDM-based methods which only use the sparse component to obtain the detection result, the low-rank part is also explored in the proposed LSDDPCRD by a density peak guided collaborative representation detector (DPCRD). Considering that there is still some anomalous information remaining in the low-rank part, using the DPCRD to process it is helpful to enhance the detection effect and performance.

Third, to utilize the information existing in HSIs comprehensively, not only the sparse matrix but also the low-rank matrix can be involved to detect anomalies. For the sake of calculating final detection results, an entropy-based adaptive fusing method is designed to combine the detection results of both low-rank and sparse parts. The method is designed based on the entropy theory to calculate the fusing weight adaptively, which can avoid the disadvantages of setting this parameter manually based on experience.

The following part of this article is arranged as follows. Section II clearly states the details of the proposed LSDDPCRD. Section III shows and analyzes experimental results. In the end, a conclusion is drawn in Section IV.

II. Proposed Method

The overall framework of the LSDDPCRD method is shown in Fig. 1. In this schematic, an HSI is first decomposed by the LSDM technique with the MoG model; then, the low-rank part is explored with the DPCRD, and the detection result of the sparse part can be obtained by adopting the Manhattan distance. Finally, with the help of the proposed entropy-based adaptive fusing method, the preliminary results of two components are fused to obtain the final detection results.

A. LSDM With the MoG Model

The technique of LSDM is on the basis of the idea that a matrix could be divided into two segments: a low-rank part and a sparse part. The basic assumption of using LSDM for hyperspectral AD is that the background of the HSI is usually low-rank while the anomaly targets are sparse in each band [36]. Due to these characteristics of HSIs and the AD task, after matrix decomposition, the low-rank part usually contains the main information of background, while the sparse part mainly has the information of anomalies [37].

Let $\boldsymbol{X} \in \boldsymbol{R}^{d\times h\times w}$ denotes an original 3-D HSI, which means that it has d spectral bands with height h and width w. The 3-D HSI is first transformed into a $d \times n$ 2-D matrix, where $n = h\times w$ denotes the number of all pixels contained in the HSI. Denote the low-rank matrix as $\boldsymbol{L} \in R^{d\times n}$ and the sparse matrix as $\boldsymbol{E} \in R^{d\times n}$, the LSDM can be mathematically formulated as

$$\min_{\boldsymbol{L},\boldsymbol{E}} \|\boldsymbol{L}\|_* + \lambda\|\boldsymbol{E}\|_1 \quad \text{s.t. } \boldsymbol{X} = \boldsymbol{L} + \boldsymbol{E}. \tag{1}$$

In the above formula, $\|\boldsymbol{L}\|_* = \sum_r \sigma_r(\boldsymbol{L})$ means the nuclear norm. $\sigma_r(L)$ is the rth singular value, where $r = 1, 2, \ldots, \min(n, d)$). $\|\cdot\|_1$ means ℓ_1-norm. The GoDec algorithm is usually used to acquire the low-rank and sparse components [38]. When the white noise is taken into consideration, formula (1) is rewritten as $\boldsymbol{X} = \boldsymbol{L} + \boldsymbol{E} + \boldsymbol{N}$, where $\boldsymbol{N}$ means the noise component. The rewritten formula is also known as stable RPCA [39].

However, the classical LSDM still has some defects and deficiencies. First, only Gaussian or Laplacian noise is considered in the model, whereas the noise in real HSIs is very complex and is difficult to estimate the noise in a single case. Second, the detection performance for the small anomalies is not very good, as these small targets may be more similar to the background and be more difficult to separate.

To solve these problems and to obtain a more accurate detection result, a novel LSDM with the MoG model (LSDM-MoG) is proposed in [37]. The MoG model has been employed

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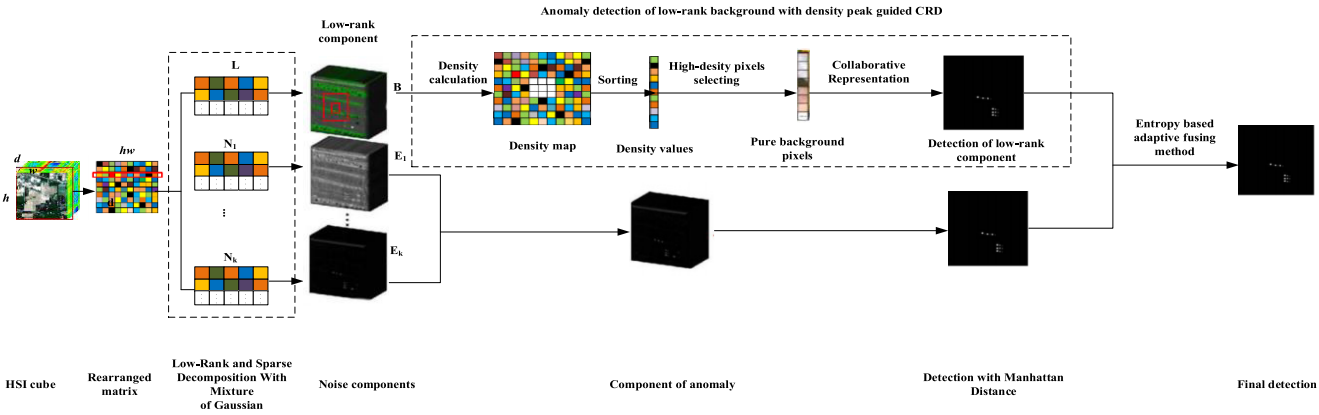


Fig. 1. Framework of the LSDDPCRD detector.

with success in the task of background or foreground separation [40]–[42]. The LSDM-MoG assumes that the MoG model can be used to estimate the complex kinds of noise in an HSI. Each e_{ij} in the sparse matrix can be represented by an MoG model as

$$e_{ij} \sim \sum_{k}^{K} \pi_k \mathrm{N}\left(e_{ij}|\mu_k, \tau_k^{-1}\right) \tag{2}$$

where $\pi_k \geq 0$ represents the proportionality parameter, $\sum_k \pi_k = 1$, K represents the number of Gaussian components, and $N\left(e|\mu, \tau^{-1}\right)$ is the Gaussian distribution with mean μ and precision τ.

The results of the LSDM-MoG are the decomposition matrices, which can be written as $\mathbf{U}$ and $\mathbf{V}$, and the reliable matrix $\mathbf{R}_k$, which has K noise components, $k = 1, 2, \ldots, K$. The following step is to separate $\mathbf{N}_k$ from the output results, where $\mathbf{N}_k$ means the complex noise components. The low-rank component can be formulated as

$$\mathbf{L} = \mathbf{U} \times \mathbf{V}^T. \tag{3}$$

Then, the reliable matrix $\mathbf{R}_k$ can be expressed as

$$\mathbf{R_k} = \boldsymbol{X} - \boldsymbol{L} = \sum_{k} \mathbf{N}_k, \quad k = 1, 2, \ldots, K. \tag{4}$$

The noise matrix $\mathbf{N}_k$ can be given as follows:

$$n_{ij} = e_{ij} \times r_{ijk} \quad \text{for } n_{ij} \in \mathbf{N}_k \tag{5}$$

where n_{ij} and r_{ijk} are the elements of the ith row and the jth column corresponding to $\mathbf{N}_k$ and $\mathbf{R}_k$. e_{ij} denotes the element of the sparse matrix.

After decomposition, the noise matrices $\mathbf{N}_k$ can be obtained. However, there are some noise in $\mathbf{N}_k$ as well as anomalous targets, and it requires to seek a reasonable component to represent anomalies. Therefore, the contrast is used to determine the component that contains the exception. The contrast c_k can be defined as

$$c_k = \max(\mathbf{N}_k) - \min(\mathbf{N}_k) \quad k = 1, 2, \ldots, K. \tag{6}$$

Compared with all the noise components, the contrast of the corresponding component of the anomalies has a maximum value. Once the component that contains anomalies is acquired, it can be converted to the original size, that is, $d \times h \times w$, and is represented as $\mathbf{E}_a$.

For the sparse component $\mathbf{E}_a$, the Manhattan distance is utilized as the detection operator gets the detection results. $\boldsymbol{e}_B$ is a standard vector which donates the background in the highest probability. In the anomaly component, as it is often sparse and approximately zero, it can be calculated as

$$e_B = \min_{e_{ij} \in \mathbf{E}_a} \left\| e_{ij} \right\|_1 \tag{7}$$

where $\boldsymbol{e}_{ij}$ is the vector of the ith row and the jth column of $\mathbf{E}_a$.

According to [37], when using Euclidean distance or Mahalanobis distance as detection operators, the inverse matrix of the covariance matrix should be calculated. However, the noise component is usually sparse after decomposition, which may cause the inverse matrix to be ill-conditioned. As there is no need to calculate the inverse of the covariance matrix, Manhattan distance is adopted.

The detection operator of the sparse matrix can be formulated as

$$r_p = \sum_{k=1}^{d} \left| e_{pk} - e_{Bk} \right| \begin{cases} \geq T, & \text{Anomaly} \\ < T, & \text{Background} \end{cases} \tag{8}$$

where r_p is the detection result of the pixel under test, e_{pk} is the kth element of the pixel under test, and T is the predefined threshold.

B. AD of Low-Rank Background With Density Peak Guided CRD

Although the LSDM-MoG model considers the existence of different kinds of noises, the anomalous information cannot be completely separated by LSDM-MoG, and the low-rank matrix may still contain some information of anomalies. Different from other traditional LSDM-based methods, to utilize the information of anomalies contained in the low-rank part comprehensively, the density peak guided CRD proposed in [43] is adopted to detect the low-rank matrix.

Let $\boldsymbol{L} \in R^{d \times h \times w}$ be the low-rank component, which is rearranged with the original size, and the density peak guided CRD is applied to process the low-rank component. In the CRD algorithm, every pixel could be represented linearly with

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the help of a background dictionary, but anomalous pixels cannot. Therefore, a background set for each pixel is usually constructed by applying the double-window strategy to select its spatial neighborhoods. Let the center pixel be γ^j, the outer window's width be ω_{out}, and the inner window's width be ω_{in}. Then, there are F pixels between the double windows, which can be expressed as

$$\gamma^j = \left\{\gamma_1^j, \gamma_2^j, \ldots, \gamma_F^j\right\}. \tag{9}$$

One drawback of this strategy is that the pixels selected as background may still contain some anomalous targets. To tackle with this problem, the density peak value is used to get a purified background set. The purified background set can be obtained in the following steps.

1) The distance vector d^j of F pixels in the background set is figured out as [44]

$$\boldsymbol{d^j} = \left\{d_1^j, d_2^j, \ldots, d_F^j\right\} \tag{10}$$

where $d_i^j = \left\|\gamma^j - \gamma_i^j\right\|_2^2$ is the Euclidean distance between γ_i^j and the pixels in dual windows.

2) The local densities $\boldsymbol{\rho}^j$ of the pixels in the background set are computed as

$$\boldsymbol{\rho}^j = \left\{\rho_1^j, \rho_2^j, \ldots, \rho_F^j\right\} \tag{11}$$

$$\boldsymbol{\rho}^j = \sum e^{-\left(\frac{D^j}{d_c^j}\right)} \tag{12}$$

where d_c denotes the cut-off distance.

3) Rearrange the elements in the obtained density vector $\boldsymbol{\rho}^j = \left\{\rho_1^j, \rho_2^j, \ldots, \rho_F^j\right\}$ in descending order, and then select m density values with larger values from high to low. The set of pixels corresponding to these values is the required pure background set $\mathbf{X}_m$. The selection method of the parameter m can be expressed by the following formula:

$$m = p \times \left(\omega_{out}^2 - \omega_{in}^2\right) \tag{13}$$

where p is a tuning parameter.

With the help of the obtained background set X_m for a pixel γ, the CRD's objective function could be given as

$$\arg\min_{\alpha} \|\gamma - \boldsymbol{X}_m\boldsymbol{\alpha}\|_2^2 + \lambda\|\boldsymbol{\alpha}\|_2^2 \tag{14}$$

where λ denotes a Lagrange multiplier. The weight vector can be figured out by deriving $\boldsymbol{\alpha}$ and setting the derivative as zero, which is

$$\hat{\boldsymbol{\alpha}} = \left(\boldsymbol{X}_m^T\boldsymbol{X}_m + \lambda\boldsymbol{I}\right)^{-1}\boldsymbol{X}_m^T\gamma. \tag{15}$$

Finally, the detection operator is

$$r_b = \|\gamma - \tilde{\gamma}\| = \|\gamma - \boldsymbol{X}_m\hat{\boldsymbol{\alpha}}\| \tag{16}$$

where γ and $\tilde{\gamma}$ denote the central pixel and its approximation separately.

C. Entropy-Based Adaptive Fusing Method

From the above analysis, to comprehensively utilize the information and detection results of the low-rank component and the sparse component, the final AD operator is determined by two parts, namely, the low-rank detection operator and the sparse detection operator. For a pixel, let r_b and r_p denote its corresponding detection results of low-rank operator and sparse operator, respectively. The final detection result can be fused as the following equation:

$$r = \omega_L^* r_b + \omega_S^* r_p \tag{17}$$

where ω_L is the weight of the low-rank component and ω_s is the weight of the sparse component, respectively.

How to set parameters r_b and r_p to get the final results becomes an urgent problem to be solved. In most cases, these parameters are set by experience or manual subjective, without considering the information of HSI. There are some drawbacks in setting these parameters subjectively because the experience and ideas of researchers are different when setting these parameters. Due to the different types of scenes and features in each HSI, the parameters for fusing the preliminary results should be set according to the characteristics and information of the image itself, and subjective or empirical setting should be avoided as far as possible. To avoid this problem and utilize the texture information and spectral information of HSIs, an entropy-based adaptive fusing method is proposed to set these weights adaptively.

As entropy is a measure of the uncertainty of the image [45], it is proposed to measure the information of the decomposed low-rank and sparse parts in the entropy-based adaptive fusing method. Let $\boldsymbol{X} = \{\boldsymbol{X_1}, \boldsymbol{X_2}, \ldots, \boldsymbol{X_d}\} \in \boldsymbol{R}^{d\times h\times w}$ be a 3-D matrix, where $\boldsymbol{X}_i \in \boldsymbol{R}^{h\times w}$ denotes the ith band of an HSI. The entropy of the ith band image is defined as [46]

$$E_i = -\sum_{g=0}^{G} p_g^{(i)} \ln p_g^{(i)} \tag{18}$$

$$p_g^{(i)} = \frac{n_g}{N} \tag{19}$$

where p_g denotes the probability that the grayscale value is g and the frequency is not 0. $N = h \times w$ denotes the number of pixels.

By summing the entropy of all band images, the total entropy of the HSI is

$$E = \frac{1}{N}\sum_{i=1}^{d} E_i. \tag{20}$$

According to the above formulas, taking the low-rank matrix and the sparse matrix as input, the entropy of the low-rank background part E_L and the entropy of the sparse part E_S can be obtained. Then, the weights of corresponding components can be calculated as

$$\omega_L = \frac{1 - E_L}{2 - (E_L + E_S)} \tag{21}$$

$$\omega_S = \frac{1 - E_S}{2 - (E_L + E_S)} \tag{22}$$

where the weight should satisfy the constraint condition [47]

$$\omega_L + \omega_S = 1. \tag{23}$$

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Therefore, the final detection operator is

$$r = \omega_L * r_b + \omega_S * r_p. \tag{24}$$

In summary, the LSDDPCRD method mainly consists of three parts, and the framework and main procedures of the LSDDPCRD are described in Algorithm 1 as pseudocodes.

Algorithm 1 The Pseudo Codes of the LSDDPCRD

Input:

Hyperspectral data $\boldsymbol{X} \in R^{d\times n}$, initial rank value l_0, the number of mixture noise K, window sizes ω_{in} and ω_{out}, the selection ratio of pure background set p.

Output:

Anomaly detection: **for** original data matrix $\boldsymbol{X}$ **do**

2: Obtain the low-rank component by using (3)

3: Obtain anomaly component by using (4), (5), and (6).

4: Obtain the detection results of the spare anomaly component by using (8)

5: Calculate the entropy and weight of anomaly component by using (20) and (22).

6: Calculate the local density by using (11).

7: Obtain the purified background set X_m of the low-rank matrix.

8: Obtain the detection of the low-rank component by using (16).

9: Calculate the entropy and weight of low-rank component by using (20) and (21).

10: **end for**

11: Calculate the final anomaly score by using (24)

III. Experimental Results and Discussion

Experiments are carried on four real HSIs to exhibit and compare the result and performance of the LSDDPCRD with other algorithms. The LSDDPCRD method is contrasted with eight classical and state-of-the-art AD algorithms, such as GRX, LRX, CRD, SRX-Segmented, RPCA-RX, LSMAD, LRASR, and LSDM-MoG. As these algorithms are typical representatives of different kinds of AD algorithms, choosing these algorithms as comparison methods is conducive to the comparison and analysis of the performance and effect of the proposed algorithm. Furthermore, the influence of some critical parameters on the LSDDPCRD is analyzed in detail.

A. Details of the Experimental Data Sets

Four real hyperspectral data sets are selected for anomalous target detection experiments. These images contain different scenarios of land-covers and were obtained by two different sensors. These data sets are representative and have been selected for experiments in many related researches and articles. Data Set1, Data Set2, and Data Set4 were obtained by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS), and Data Set3 was captured by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor. Fig. 2 gives the pseudocolor images and ground truth maps of these data sets. The first two data sets are selected from the same HSI, which shows the image of the San Diego airport, USA. It has 126 bands with the spatial resolution of 3.5 m. Take an image with a length of 100 and a width of 100 from the original image as the first data set [48]. Then, take an image with a length of 90 and a width of 100 from the original image as the second data set [49]. The other two data sets are selected from different scenes in the Airport-Beach-Urban (ABU) data set [50]. The size of the third data set is 150×150. This image has 102 spectral bands with a spatial resolution of 1.3 m. The size of the fourth HSI is 100×100, which contains 191 spectral bands with the spatial resolution of 3.4 m.

B. Evaluation Criteria and Experimental Settings

The evaluation criteria, compared algorithms, and the parameters of the LSDDPCRD are introduced in this section.

1) Evaluation Criteria: In hyperspectral AD, the receiver operating characteristics (ROC) curve is adopted in this article to exhibit detectors' performance directly. As the ROC curve cannot reflect the performance of a detector quantitatively, the area under the ROC curve (AUC) is also chosen for evaluation. These two indicators are selected as evaluation criteria in many related research articles [48]–[50]. There are a few terms that should be known before introducing the ROC curve. The first one is the probability of detection rate, P_d. The other is the probability of false alarm, P_f. A successful AD has high P_d and low P_f. P_d and P_f can be expressed in the following formula:

$$\begin{cases} P_d = \dfrac{N_d}{N_t} \\ P_f = \dfrac{N_{\text{miss}}}{N_t} \end{cases} \tag{25}$$

where N_d is the number of anomalous targets also detected as anomalous pixels correctly, N_t is defined as the total number of anomalous targets in the actual image, and N_{miss} is defined as the total number of anomalous targets in the image but detected as background pixels and background pixels in the image but detected as anomalous targets.

The ROC curve indicates the relationship between P_d and P_f [48], where P_f is the horizontal axis and P_d is the vertical axis. Generally speaking, the curve close to the upper left corner means that the performance of a detector is good. When it is not possible to directly see which algorithm has better detection performance through the ROC curve, the area under the ROC curve is often chosen as another important criteria to judge the performance of the algorithm. Compared with the intuitiveness of the ROC, the AUC value can quantitatively evaluate the accuracy of detection algorithm. The larger the AUC value, which is closer to 1, the better the detector's performance.

2) Compared Methods: To show and reveal the results of the LSDDPCRD algorithm, eight classical and state-of-the-art detectors are selected for comparison. These compared algorithms consist of GRX [13], LRX [15], CRD [20], SRX-Segmented [19], RPCA-RX [28], LSMAD [35], LRASR [21], and the LSDM-MoG method [37]. These eight algorithms are selected by taking the representativeness and novelty of the algorithms into consideration. GRX and LRX

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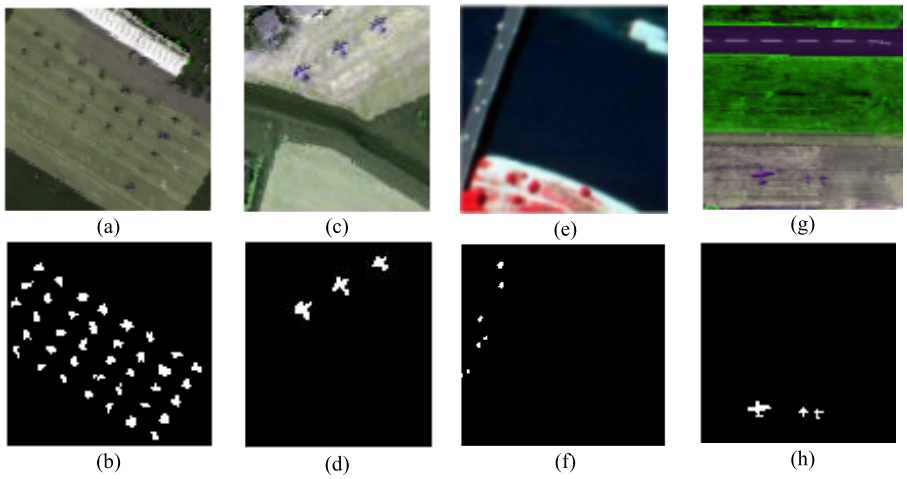


Fig. 2. Pseudocolor of HSIs and their ground truth maps. (a) and (b) Data Set1. (c) and (d) Data Set2. (e) and (f) Data Set3. (g) and (h) Data Set4.

algorithms are classical algorithms based on background statistical theory, which are the baseline of the AD area and have been selected for comparison nearly in all related articles. SRX-Segmented and RPCA-RX are two improved versions based on RX algorithm, and RPCA-RX also uses the strategy of LSDM. The CRD algorithm can be seen as a typical representation theory-based method. It does not need to estimate the background statistically, and usually has a good performance for detection.

Since the proposed algorithm uses the idea of LSDM to separate anomalies from the background, some related algorithms are selected. The LSMAD was proposed by Sun *et al.* [35]. The LRASR is a popular detector for HSIs based on low-rank and sparse representation [21]. As mentioned in [21] and [35], these two algorithms have very good performance. In addition, the proposed LSDDPCRD algorithm takes the mixture of noise into account, which is more consistent with the true hyperspectral data. As a result, the LSDM-MoG is also added to the comparison experiment [37].

In summary, a variety of classical and popular algorithms are selected for comparison, and these detectors were designed based on different strategies for the AD task. Therefore, the selection of comparison algorithm is reasonable and can reflect the performance of the proposed detector comprehensively.

3) Settings of Parameters: Table I illustrates selected parameters for all algorithms used for experiments. LRX has two parameters: inner window size ω_{in} and outer window size ω_{out}. LSMAD has two parameters: maximum rank of a low-rank matrix r and cardinality of a sparse matrix k. LRASR has four parameters: clustering number of k-means algorithm K, the number of pixels constituting the minimum Mahalanobis distance of the dictionary P, the equilibrium parameter between sparse matrix and coefficient matrix λ, and the parameter β, which is used to adjust the low rankness and sparsity. LSDM-MoG has two parameters: initial rank l_0 and number of noise k. There are three parameters in CRD: ω_{in}, ω_{out}, and regularization parameter λ. LSDDPCRD has five parameters: initial rank l_0, number of noise k, ω_{in}, ω_{out}, and proportion of pure background set selection p. In Section III-D, the parameter selection of the proposed algorithm and the influence of different parameters on the experimental results are discussed in detail.

C. Performance Comparison Between LSDDPCRD and Other AD Algorithms

Figs. 3–6 depict the detection results obtained by using different AD algorithms on four HSI data sets. From Figs. 3 to 6, it is easy to find that some comparison algorithms can only detect a small number of anomalous targets. From the detection result maps of some compared methods on the four data sets, it is difficult to find all anomalous targets. For example, in Fig. 3, the LRX method can only find three anomalies, but it detected many background pixels as anomalies. The methods of GRX, RPCA-RX, and LSMAD also have very poor detection performance on the first image. Among these compared methods, the LSDM-MoG achieved the best performance on the first data set. In sharp contrast to these algorithms, the LSDDPCRD can detect more anomalous pixels in the image. By analyzing and comparing the results of each graph, we can see that the effect of methods based on statistical hypothesis is always general or poor, such as GRX and LRX. This may be due to the complexity of environment composition and data distribution of an HSI, which cannot be estimated by a simple data distribution. A remarkable feature of the CRD is that the double window strategy is adopted, which can take advantage of the piecewise smoothing characteristics of HSIs and reduce the possibility of abnormal pixels mixing into the background set by using spatial information, so as to improve the detection effect and detection ability. From the experimental results, it can be concluded that the CRD method can detect anomalies, but it suffers from the problem of high false alarm, for example, the results shown in Fig. 5. Through further analysis of Figs. 3–6, it can be found that the LSDDPCRD algorithm can detect anomalous targets more accurately. The reason for this phenomenon might be the consideration of many kinds of noise, along with using of local density information to help further purify the background pixels.

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TABLE I
CHOSEN PARAMETERS OF ALL ALGORITHMS

Method	Parameter	Dataset1	Dataset2	Dataset3	Dataset4
LRX [15]	ω_{in}	3	5	3	3
	ω_{out}	5	7	5	5
LSMAD [35]	r	11	12	31	23
	k	10^4	10^5	10^5	10^5
LRSAR [21]	K	15	15	15	15
	P	20	20	20	20
	λ	0.1	0.1	0.1	0.1
	β	0.1	0.1	0.1	0.1
LSDM-MoG [37]	l_0	60	70	10	10
	K	1	1	3	2
CRD [20]	ω_{in}	3	17	3	3
	ω_{out}	15	19	5	9
	λ	10^{-6}	10^{-6}	10^{-6}	10^{-6}
LSDDPCRD (ours)	l_0	60	70	10	10
	K	1	1	3	2
	ω_{in}	5	5	3	3
	ω_{out}	15	23	9	13
	p	0.1	0.2 or 0.3	0.7	0.2

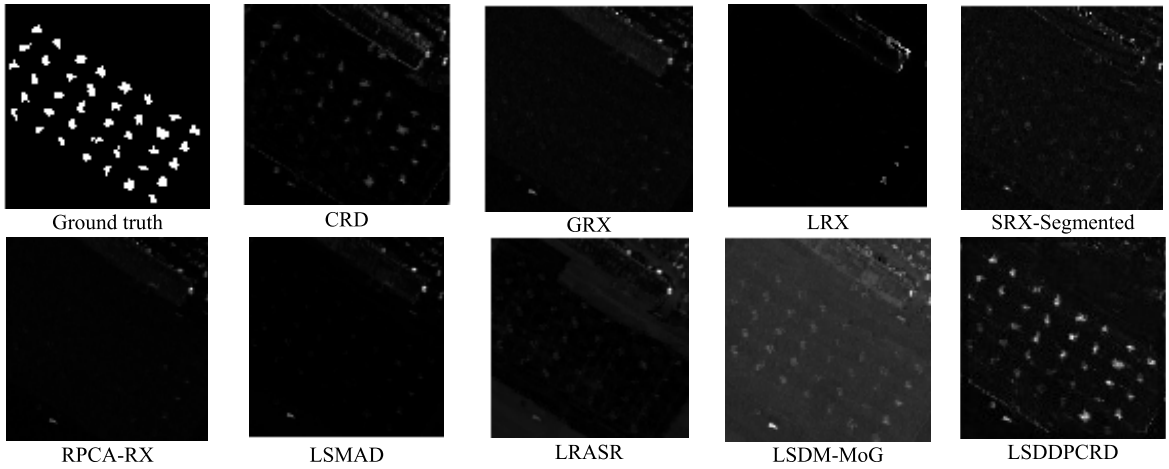


Fig. 3. Detection maps of all detectors on Data Set1.

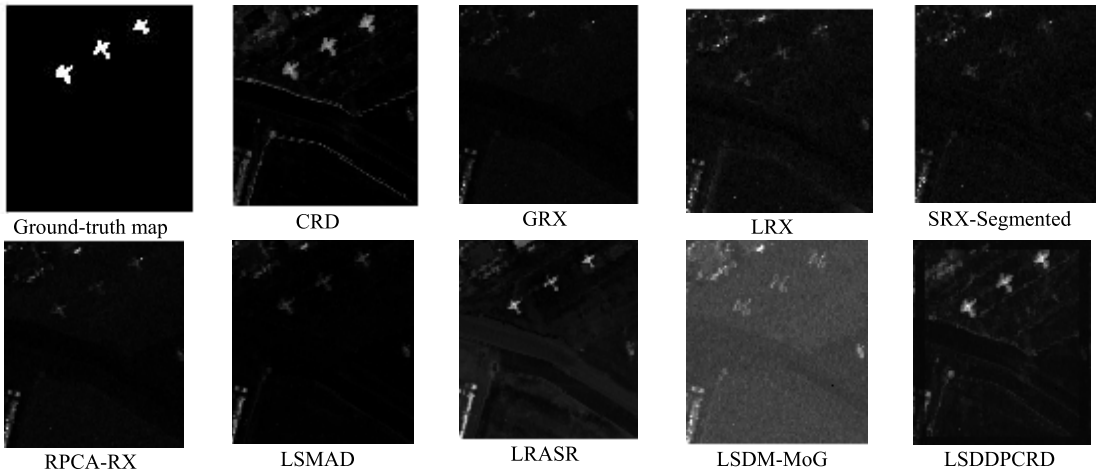


Fig. 4. Detection maps of all detectors on Data Set2.

As the influence of the noise problem is reduced and the possibility of mixing the information of abnormal targets into the background set is reduced, higher detection accuracy can be acquired. In addition, it is not difficult to find in the detection graph that the shapes of the anomalies detected by some compared detectors are destroyed. For example, in Fig. 6,

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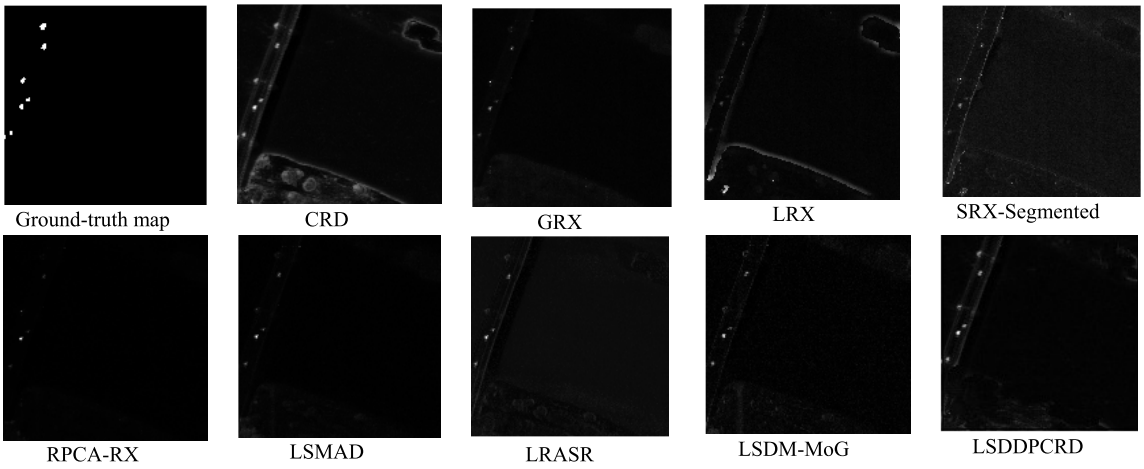


Fig. 5. Detection maps of all detectors on Data Set3.

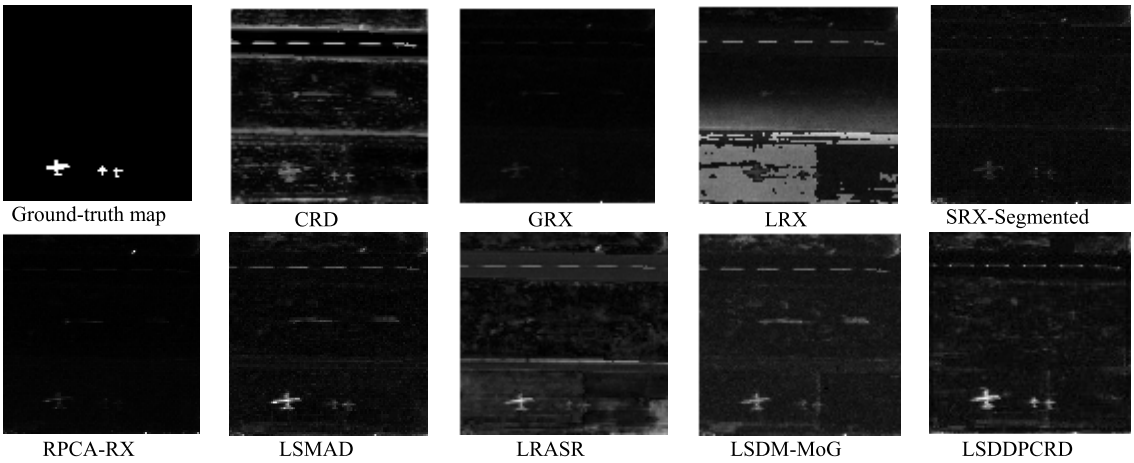


Fig. 6. Detection maps of all detectors on Data Set4.

the shapes of anomalous targets detected by LRASR and LSDM-MoG are not very good, when compared with the corresponding ground-truth map. Conversely, the LSDDPCRD algorithm can detect most anomalous objects effectively while retaining the shapes of the anomalies. The reason why the LSDDPCRD method achieves good results may be that the proposed method considers the existence of multiple kinds of noises, and comprehensively uses the information of low-rank component and sparse component for AD.

Fig. 7 depicts the ROC curves of all these algorithms on four data sets. From Fig. 7(a), it is obvious that when the false alarm rate changes from 0 to 1, the curve of the LSDDPCRD algorithm is always higher than the curves obtained by other algorithms, which means that the performance of the LSDDPCRD is higher than other detectors used for comparison. From Fig. 7(b) to (d), it could be found that the curve of the LSDDPCRD also locates at the left corner of the graph, which indicates that the LSDDPCRD performs better than other detectors used in the experiments. To further address and compare the performance of these algorithms on four HSI data sets quantitatively, the AUC scores of all these detectors are provided in Table II. As illustrated in Table II, it is obvious that the detection accuracy of different scenes acquired by different hyperspectral sensors is improved by LSDDPCRD. The proposed LSDDPCRD method has the best performance among these algorithms, whose AUC scores on four data sets are 0.9409, 0.9979, 0.9882, and 0.9974, respectively. Therefore, based on the AUC values obtained by each algorithm given in Table II, we can still conclude that the detection performance of the LSDDPCRD algorithm is better when contrasted with other algorithms.

To analyze and discuss the performance of all detectors from another point of view, the separability maps of the LSDDPCRD with several all compared methods are plotted in Fig. 8. The degree of separation of anomalous targets and background pixels can be used to show that the performance of a detection method. If the degree of separation is large, the anomalies can be detected easily from the background, which indicates that the detector has a good performance. In the graph, red donates the anomalous target, and green donates the background, which can show the distribution of background and anomalies. Their position proves that the background and anomalies can be separated. As for Data Set1, Data Set3 and Data Set4 shown in Fig. 8(a), (c), and (d), it is obvious that the distance between the anomalies and the background of the proposed algorithm LSDDPCRD is larger than that of other algorithms. For Data Set 2, the distance between the anomalies and the background of the LSDDPCRD

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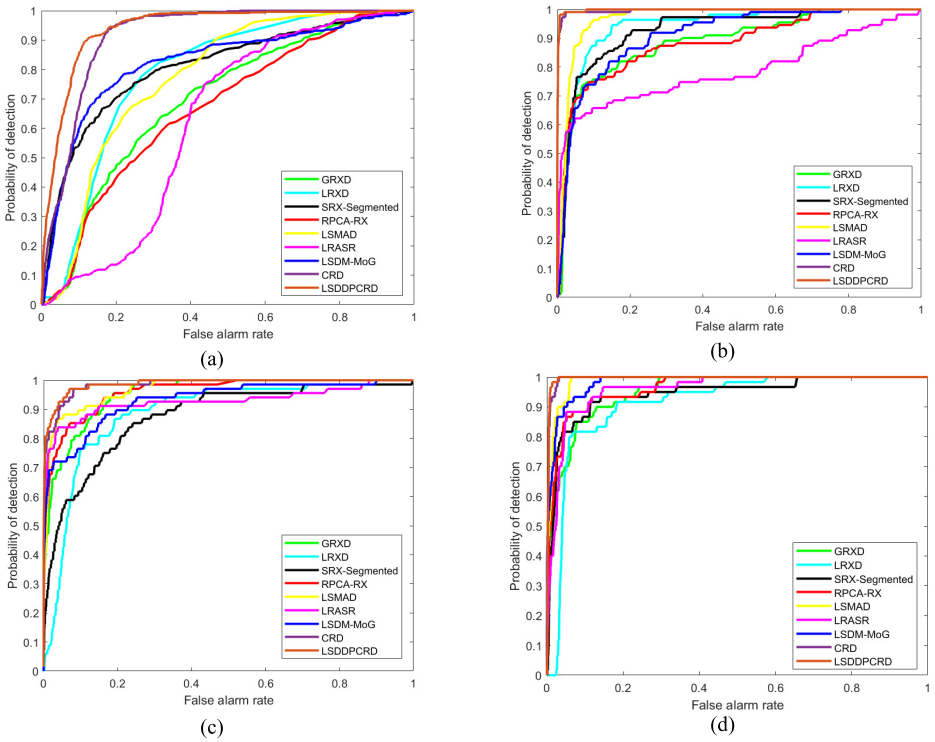


Fig. 7. ROC curves of nine detectors on four data sets. (a) Data Set1. (b) Data Set2. (c) Data Set3. (d) Data Set4.

TABLE II
AUC SCORES OF NINE DETECTORS ON FOUR DATA SETS

Dataset1		Dataset2		Dataset3		Dataset4	
Method	AUC	Method	AUC	Method	AUC	Method	AUC
CRD [20]	0.9169	CRD	0.9963	CRD	0.9855	CRD	0.8217
GRX [13]	0.6983	GRX	0.8906	GRX	0.9538	GRX	0.9526
LRX [15]	0.7881	LRX	0.9411	LRX	0.8868	LRX	0.6528
SRX-Segmented [19]	0.8091	SRX-Segmented	0.9275	SRX-Segmented	0.8758	SRX-Segmented	0.9445
RPCA-RX [28]	0.6705	RPCA-RX	0.8845	RPCA-RX	0.9599	RPCA-RX	0.9627
LSMAD [35]	0.7739	LSMAD	0.9690	LSMAD	0.9698	LSMAD	0.9878
LRSAR [21]	0.6228	LRASR	0.7876	LRASR	0.9239	LRASR	0.9535
LSDM-MoG [37]	0.8229	LSDM-MoG	0.9107	LSDM-MoG	0.9251	LSDM-MoG	0.9605
LSDDPCRD (ours)	**0.9409**	LSDDPCRD	**0.9979**	LSDDPCRD	**0.9882**	LSDDPCRD	**0.9974**

algorithm is similar to that of the CRD, and larger than those of other algorithms. Therefore, the separability of background and anomalies of different algorithms shows that the anomalies detect by the proposed algorithm can be easier separated from the background than those detected by other detectors, from which the effectiveness and superiority of the algorithm are illustrated.

By analyzing the above experiments, it can be seen that all results indicate that the proposed detector has good detection performance and effect. There might be some reasons for the good results and effects of the proposed LSDDPCRD method. First, the LSDM theory combined with the MoG model is first taken to process the original HSI, which can alleviate the influence of the noise problem on the detection results. Second, the density peak guided CRD is adopted to explore the low-rank part, which can take advantage of the spatial information and enhance the detector's effect and performance. Third, an entropy-based adaptive fusing method is designed to fuse the detection results of both the low-rank operator and the sparse operator, which could utilize the information existing in HSIs comprehensively.

D. Parametric Analysis

To further illustrate the performance of the proposed LSDDPCRD, the influence of some important parameters associated with the proposed algorithm on its results and performance is analyzed in this section. The parameters of the LSDDPCRD detector mainly include three parts: the parameters associated with the LSDM-MoG part, the parameters associated with the DPCRD, and the parameters associated with the entropy-based adaptive fusing method.

The parameters associated with the LSDM-MoG part are l_0, which is the initial rank, and K, which is the initial number of noise. The influence of these parameters has been discussed in detail, which can be found in [37]. Therefore, the influence of these parameters on the results of LSDM will not be discussed repeatedly in this article. The parameters related to the

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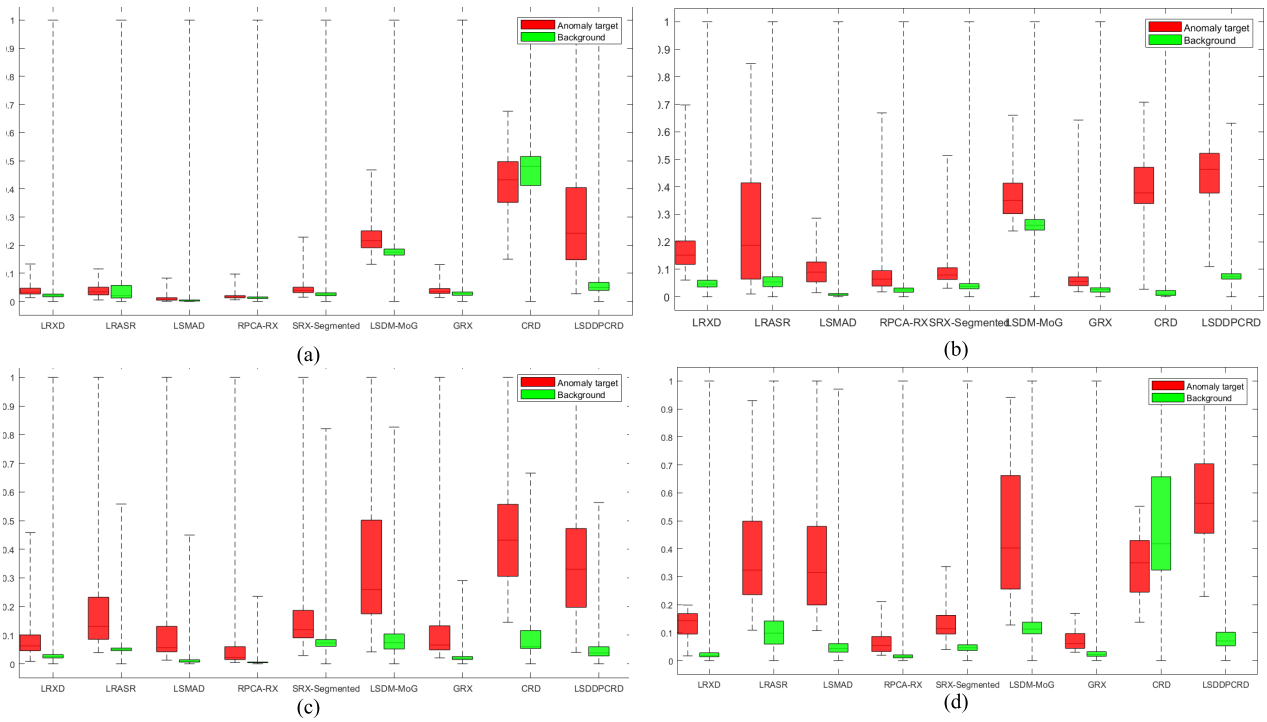


Fig. 8. Separability maps of anomaly detectors on four data sets. (a) Data Set1. (b) Data Set2. (c) Data Set3. (d)Data Set4.

LSDM-MoG part are set by referring to the recommendations given in [37]. In detail, the initial rank of the LSDDPCRD method for four data sets is set as 60, 70, 10, 10, and the LSDDPCRD method's initial number K is set as 1, 1, 3, 2, respectively.

This section mainly analyzes the influence of parameters associated with the DPCRD and the parameters associated with the entropy-based adaptive fusing method on the results and performance of the LSDDPCRD algorithm. The parameters of DPCRD are mainly the selected dual window sizes (ω_{out} and ω_{in}), and the parameter p. The influence of the sizes of dual windows is analyzed first. Table III shows the AUC values obtained by the proposed algorithm when using different dual window sizes on four data sets. As illustrated in Table III, when window sizes ($\omega_{\text{out}}, \omega_{\text{in}}$) for the four data sets are set to (15, 5), (23, 5), (9, 3), (13, 3), the highest AUC values of the LSDDPCRD algorithm can be obtained. Moreover, with the change of a series of window sizes, the lowest AUC score on the first data set is 0.9286, and the highest AUC score is 0.9339. For the rest data sets, their responding lowest and highest AUC scores are (0.9965, 0.9972), (0.9741, 0.9863), and (0.9924, 0.9973), respectively. It can be seen that under different window sizes, the AUC scores of the proposed LSDDPCRD algorithm have changed, but the change of the LSDDPCRD algorithm's AUC value on all data sets is not so obvious. The phenomenon may imply that the density peak guided strategy can purify the background and improve the algorithm's performance.

To further illustrate the effect of the density peak guided strategy, the influence of the parameter p is discussed. Although the double window strategy can avoid the anomalous pixels from mixing into the background pixel set to a certain extent, due to the irregular shape of anomalous targets and the different selection of the double window scale, a small number of anomalous pixels may still exist in the background set selected by the double window strategy. Therefore, density peak guided strategy can be used to eliminate the anomalies mixed into the background set. The parameter p is used to determine how many pixels are finally selected between the dual windows to form the background set. Table IV gives the AUC scores of the LSDDPCRD method for four data sets with different values of the parameter p. A reasonable parameter p can maximize the retention of the elements closest to the center pixel, and can make the selected background set purer, which further enhance the detection performance. From Table IV, it is easy to find that the LSDDPCRD can acquire the highest AUC score, which are 0.9409, 0.9871, and 0.9977, when the parameter p for Data Set1, Data Set3, and Data Set4 are set to 0.1, 0.7, and 0.2, respectively. When the parameter p for Data Set2 is 0.2 or 0.3, the AUC score of the proposed method is 0.9979.

Finally, whether the entropy-based adaptive fusing method can set the optimal weights (ω_L, ω_S) for fusing the detection results of the low-rank operator and the sparse operator is analyzed. Due to the requirement that the sum of ω_L and ω_S should be 1, if the value of ω_L is given, the other weight is also set. By adjusting the value of weight ω_L from 0.1 to 1 by step 0.1 for four hyperspectral data sets, the corresponding AUC scores of the proposed method can be obtained, which are given in Table V. With the increase of the value of ω_L, the AUC scores of the LSDDPCRD method increase first and then decrease on Data Set1 and Data Set2. For Data Set3 and Data Set4, although the AUC scores fluctuate slightly, the general trend is still gradually reaching the maximum value and then

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TABLE III
AUC SCORES OF THE LSDDPCRD WITH DIFFERENT WINDOW SIZE ($\omega_{\text{OUT}}, \omega_{\text{IN}}$) ON DIFFERENT DATA SETS

Dataset1			Dataset2			Dataset3			Dataset4		
ω_{out}	ω_{in}	AUC	ω_{out}	ω_{in}	AUC	ω_{out}	ω_{in}	AUC	ω_{out}	ω_{in}	AUC
15	3	0.9327	21	3	0.9965	5	3	0.9741	11	3	0.9955
11	5	0.9286	23	3	0.9970	7	3	0.9816	**13**	**3**	**0.9973**
13	5	0.9318	25	3	0.9968	**9**	**3**	**0.9863**	15	3	0.9964
15	**5**	**0.9339**	21	5	0.9967	11	3	0.9856	11	5	0.9967
17	5	0.9336	**23**	**5**	**0.9972**	13	3	0.9843	13	5	0.9970
19	5	0.9322	25	5	0.9970	15	3	0.9843	15	5	0.9924

TABLE IV
AUC SCORES OF THE LSDDPCRD WITH DIFFERENT p ON FOUR DATA SETS

Dataset1		Dataset2		Dataset3		Dataset4	
p	AUC	p	AUC	p	AUC	p	AUC
0.1	**0.9409**	0.1	0.9976	0.1	0.9803	0.1	0.9971
0.2	0.9376	**0.2**	**0.9979**	0.2	0.9807	**0.2**	**0.9977**
0.3	0.9359	**0.3**	**0.9979**	0.3	0.9773	0.3	0.9958
0.4	0.9349	0.4	0.9978	0.4	0.9823	0.4	0.9971
0.5	0.9343	0.5	0.9976	0.5	0.9803	0.5	0.9957
0.6	0.9341	0.6	0.9974	0.6	0.9796	0.6	0.9951
0.7	0.9339	0.7	0.9973	**0.7**	**0.9871**	0.7	0.9941
0.8	0.9339	0.8	0.9972	0.8	0.9863	0.8	0.9973
0.9	0.9339	0.9	0.9970	0.9	0.9863	0.9	0.9969
1	0.9339	1	0.9968	1	0.9863	1	0.9962

TABLE V
AUC SCORES OF THE LSDDPCRD WITH DIFFERENT ω ON FOUR DATA SETS

Dataset1		Dataset2		Dataset3		Dataset4	
ω	AUC	ω	AUC	ω	AUC	ω	AUC
0.1	0.9183	0.1	0.9888	0.1	0.9557	0.1	0.9924
0.2	0.9347	0.2	0.9949	0.2	0.9770	0.2	0.9927
0.3	0.9396	0.3	0.9970	0.3	0.9829	0.3	0.9925
0.4	**0.9411**	0.4	0.9977	0.4	0.9752	0.4	0.9972
0.5	0.9409	**0.5**	**0.9979**	0.5	0.9871	**0.5**	**0.9977**
0.6	0.9393	0.6	0.9978	**0.6**	**0.9888**	0.6	0.9975
0.7	0.9369	0.7	0.9978	0.7	0.9876	0.7	0.9964
0.8	0.9323	0.8	0.9977	0.8	0.9860	0.8	0.9962
0.9	0.9285	0.9	0.9976	0.9	0.9857	0.9	0.9957
1	0.9248	1	0.9975	1	0.9859	1	0.9955

decreasing. These results show that it is effective to get the final result by fusing the results of both the low-rank operator and the sparse operator at the same time. Furthermore, from Table V, it is easy to find that the optimal weights (ω_L, ω_S) of the four data sets are (0.4, 0.6), (0.5, 0.5), (0.6, 0.4), and (0.5, 0.5) respectively, and their corresponding AUC scores of the LSDDPCRD method are 0.9411, 0.9979, 0.9888, and 0.9977, respectively. As a contrast, the weights set of the four data sets by the entropy-based adaptive fusing method are (0.5054, 0.4946), (0.5034, 0.4966), (0.5492, 0.4508), and (0.5092, 0.4908), respectively, and the corresponding AUC scores are 0.9409, 0.9979, 0.9882, and 0.9974, respectively. Through comparison and analysis, it can be shown that the weights set by the proposed entropy-based adaptive fusing method are nearly the same as the optimal ones, which shows that the proposed entropy-based adaptive fusing method can effectively obtain reasonable weights while ensuring detection performance.

IV. CONCLUSION

A detector based on LSDDPCRD was proposed for HSIs in this article. Different from traditional LSDM-based AD methods, the LSDDPCRD algorithm makes use of both the sparse component and the low-rank background component to detect anomalies. First, the original hyperspectral data are decomposed by the LSMD-MoG model to get the low-rank part and sparse component. By utilizing the MoG model to consider multiple kinds of noise existing in the real environment, the problem of background noise contamination can be alleviated. Then, for the sparse matrix, the Manhattan distance

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was selected as the detection operator. As there is still some anomalous information remaining in the low-rank matrix, DPCRD is used to process the low-rank matrix. Finally, to get the final detection results an entropy-based adaptive fusing method is designed to fuse the results obtained from the low-rank matrix and sparse matrix, which can calculate the fusing weights adaptively according to the characteristics of an HSI. By verifying the LSDDPCRD algorithm and other classic and recently popular methods on four real HSI data sets, it is not difficult to conclude that the LSDDPCRD algorithm has better detection performance and effectiveness.

REFERENCES

[1] C. Zhao, C. Li, S. Feng, N. Su, and W. Li, "A spectral–spatial anomaly target detection method based on fractional Fourier transform and saliency weighted collaborative representation for hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 5982–5997, 2020.

[2] Y. Wang, L. Wang, H. Xie, and C.-I. Chang, "Fusion of various band selection methods for hyperspectral imagery," *Remote Sens.*, vol. 11, no. 18, p. 2125, Sep. 2019.

[3] L. Ren, L. Zhao, and Y. Wang, "A superpixel-based dual window RX for hyperspectral anomaly detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 7, pp. 1233–1237, Jul. 2020.

[4] M. Zhang, W. Li, and Q. Du, "Diverse region-based CNN for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 27, no. 6, pp. 2623–2634, Jun. 2018.

[5] Y. Lu and S. Huang, "Sparse representation based hyperspectral anomaly detection via adaptive background sub-dictionaries," *IEEE Access*, early access, Oct. 29, 2020, doi: 10.1109/ACCESS.2020.3034796.

[6] Y. Wang *et al.*, "A posteriori hyperspectral anomaly detection for unlabeled classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3091–3106, Jun. 2018.

[7] X. Xu, J. Li, and S. Li, "Multiview intensity-based active learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 669–680, Feb. 2018.

[8] J. Li, H. Zhang, L. Zhang, and L. Ma, "Hyperspectral anomaly detection by the use of background joint sparse representation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2523–2533, Jun. 2015.

[9] Y. Wang *et al.*, "Constrained-target band selection for multiple-target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 6079–6103, Aug. 2019.

[10] Z. Li and Y. Zhang, "Hyperspectral anomaly detection based on improved RX with CNN framework," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2019, pp. 2244–2247.

[11] B. Cheng and L. Zhang, "Optimal band subspace hyperspectral anomaly detection based on bilateral filtering," *Liquid Cryst. Display*, vol. 34, no. 9, pp. 897–904, 2019.

[12] S. Matteoli, M. Diani, and G. Corsini, "A tutorial overview of anomaly detection in hyperspectral images," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 25, no. 7, pp. 5–28, Jul. 2010.

[13] I. S. Reed and X. Yu, "Adaptive multiple-band CFAR detection of an optical pattern with unknown spectral distribution," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 38, no. 10, pp. 1760–1770, Oct. 1990.

[14] D. Manolakis and G. Shaw, "Detection algorithms for hyperspectral imaging applications," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 29–43, Jan. 2002.

[15] R. Zhao, B. Du, and L. Zhang, "Hyperspectral anomaly detection via a sparsity score estimation framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 6, pp. 3208–3222, Jun. 2017.

[16] H. Kwon and N. M. Nasrabadi, "Kernel RX-algorithm: A nonlinear anomaly detector for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 2, pp. 388–397, Feb. 2005.

[17] W.-M. Liu and C.-I. Chang, "Multiple-window anomaly detection for hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 2, pp. 644–658, Apr. 2013.

[18] W. Li and Q. Du, "Decision fusion for dual-window-based hyperspectral anomaly detector," *J. Appl. Remote Sens.*, vol. 9, no. 1, Feb. 2015, Art. no. 097297.

[19] N. Gorelnik, H. Yehudai, and S. R. Rotman, "Anomaly detection in non-stationary backgrounds," in *Proc. 2nd Workshop Hyperspectral Image Signal Process., Evol. Remote Sens.*, Jun. 2010, pp. 1–4.

[20] W. Li and Q. Du, "Collaborative representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1463–1474, Mar. 2015.

[21] Y. Xu, Z. Wu, J. Li, A. Plaza, and Z. Wei, "Anomaly detection in hyperspectral images based on low-rank and sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 4, pp. 1990–2000, Apr. 2016.

[22] X. Zhang, G. Wen, and W. Dai, "A tensor decomposition-based anomaly detection algorithm for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 5801–5820, Oct. 2016.

[23] N. Li *et al.*, "Anomaly detection based on extended mathematical morphology for hyperspectral imagery," *Acta Optica Sinica*, vol. 28, no. 8, pp. 1480–1484, 2008.

[24] J. Liao and L. Wang, "Hyperspectral Image classification method based on fusion with two kinds of spatial information," *Laser Optoelectron. Prog.*, vol. 54, no. 8, p. 081002, 2017.

[25] K. Tan, Z. Hou, F. Wu, Q. Du, and Y. Chen, "Anomaly detection for hyperspectral imagery based on the regularized subspace method and collaborative representation," *Remote Sens.*, vol. 11, no. 11, p. 1318, Jun. 2019.

[26] Z. Yuan, H. Sun, K. Ji, Z. Li, and H. Zou, "Local sparsity divergence for hyperspectral anomaly detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 10, pp. 1697–1701, Oct. 2014.

[27] L. Du, Z. Wu, Y. Xu, W. Liu, and Z. Wei, "Kernel low-rank representation for hyperspectral image classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 477–480.

[28] W. Sun, C. Liu, J. Li, Y. M. Lai, and W. Li, "Low-rank and sparse matrix decomposition-based anomaly detection for hyperspectral imagery," *J. Appl. Remote Sens.*, vol. 8, no. 1, May 2014, Art. no. 083641.

[29] X. Cui, Y. Tian, L. Weng, and Y. Yang, "Anomaly detection in hyperspectral imagery based on low-rank and sparse decomposition," *Proc. SPIE*, vol. 9069, Jan. 2014, Art. no. 90690R.

[30] W. Sun, L. Tian, Y. Xu, B. Du, and Q. Du, "A randomized subspace learning based anomaly detector for hyperspectral imagery," *Remote Sens.*, vol. 10, no. 3, p. 417, Mar. 2018.

[31] E. J. Candès, X. Li, Y. Ma, and J. Wright, "Robust principal component analysis?" *J. ACM*, vol. 58, no. 3, p. 11, May 2011.

[32] N. Vaswani, T. Bouwmans, S. Javed, and P. Narayanamurthy, "Robust subspace learning: Robust PCA, robust subspace tracking, and robust subspace recovery," *IEEE Signal Process. Mag.*, vol. 35, no. 4, pp. 32–55, Jul. 2018.

[33] T. Bouwmans, A. Sobral, S. Javed, S. K. Jung, and E.-H. Zahzah, "Decomposition into low-rank plus additive matrices for background/foreground separation: A review for a comparative evaluation with a large-scale dataset," *Comput. Sci. Rev.*, vol. 23, pp. 1–71, Feb. 2017.

[34] W. Sun, G. Yang, J. Li, and D. Zhang, "Randomized subspace-based robust principal component analysis for hyperspectral anomaly detection," *J. Appl. Remote Sens.*, vol. 12, no. 1, p. 1, Mar. 2018.

[35] Y. Zhang, B. Du, L. Zhang, and S. Wang, "A low-rank and sparse matrix decomposition-based mahalanobis distance method for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1376–1389, Mar. 2016.

[36] F. Küçük, B. Töreyin, and F. V. Çelebi, "Sparse and low-rank matrix decomposition-based method for hyperspectral anomaly detection," *J. Appl. Remote Sens.*, vol. 13, no. 1, p. 014519, Feb. 2019.

[37] L. Li, W. Li, Q. Du, and R. Tao, "Low-rank and sparse decomposition with mixture of Gaussian for hyperspectral anomaly detection," *IEEE Trans. Cybern.*, early access, Feb. 25, 2020, doi: 10.1109/TCYB.2020.2968750.

[38] C.-I. Chang, H. Cao, S. Chen, X. Shang, C. Yu, and M. Song, "Orthogonal subspace projection-based go-decomposition approach to finding low-rank and sparsity matrices for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, early access, Jul. 14, 2020, doi: 10.1109/TGRS.2020.3002724.

[39] Z. Zhou, X. Li, J. Wright, E. Candès, and Y. Ma, "Stable principal component pursuit," in *Proc. IEEE Int. Symp. Inf. Theory*, Austin, TX, USA, Jun. 2010, pp. 1518–1522.

[40] X. Cao *et al.*, "Low-rank matrix factorization under general mixture noise distributions," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1493–1501.

[41] H. Yong, D. Meng, W. Zuo, and L. Zhang, "Robust online matrix factorization for dynamic background subtraction," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 7, pp. 1726–1740, Jul. 2018.

[42] C. Liu, C. Chen, H. Shan, and B. Wang, "Robust online matrix completion with Gaussian mixture model," in *Proc. ICASSP - IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Barcelona, Spain, May 2020, pp. 3422–3426.

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[43] B. Tu, N. Li, Z. Liao, X. Ou, and G. Zhang, "Hyperspectral anomaly detection via spatial density background purification," *Remote Sens.*, vol. 11, no. 22, p. 2618, Nov. 2019.
[44] A. Rodriguez and A. Laio, "Clustering by fast search and find of density peaks," *Science*, vol. 344, no. 6191, pp. 1492–1496, Jun. 2014.
[45] M. Xia and Z. Xu, "Entropy/cross entropy-based group decision making under intuitionistic fuzzy environment," *Inf. Fusion*, vol. 13, no. 1, pp. 31–47, Jan. 2012.
[46] S. Piros and P. Korondi, "Entropy based adaptive image compression procedure," in *Proc. Adv. Sci. Eng. Technol. Int. Conf. (ASET)*, Feb. 2018, pp. 1–5.
[47] Y. Li, X. Shi, and Q. Zhao, "Hyperspectral imaging classification based on band weighted K-means clustering," *Radio Eng.*, vol. 50, no. 11, pp. 911–916, 2020.
[48] S. Khazai and B. Mojaradi, "A modified kernel-RX algorithm for anomaly detection in hyperspectral images," *Arabian J. Geosci.*, vol. 8, no. 3, pp. 1487–1495, Mar. 2015.
[49] G. Zhang, M. Xu, Y. Zhang, and Y. Fan, "Improved hyperspectral anomaly target detection method based on mean value adjustment," in *Proc. 10th Workshop Hyperspectral Imag. Signal Process., Evol. Remote Sens. (WHISPERS)*, Amsterdam, The Netherlands, Sep. 2019, pp. 1–4.
[50] X. Kang, X. Zhang, S. Li, K. Li, J. Li, and J. Atli Benediktsson, "Hyperspectral anomaly detection with attribute and edge-preserving filters," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 10, pp. 5600–5611, Oct. 2017.



Shou Feng (Member, IEEE) is an Assistant Professor with Harbin Engineering University, Harbin, China.

His main research interest includes hyperspectral image processing and data mining.



Shulu Tang is currently pursuing the master's degree with Harbin Engineering University, Harbin, China.

Her main research areas are hyperspectral image processing.



Chunhui Zhao is working as a Professor and a Doctoral Supervisor with the College of Information and Communication Engineering, Harbin Engineering University, Harbin, China.

His research interests include mathematical morphology and hyperspectral image processing.



Ying Cui is working as an Associate Professor with Harbin Engineering University, Harbin, China.

Her main research interest includes remote sensing image processing.

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